

Amirarash (Aaron) Kashef, Ph.D.

Machine Learning Engineer | Data Scientist | AI Researcher

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PROFESSIONAL SUMMARY

ML Engineer and Data Scientist with 4+ years of end-to-end experience building and deploying multimodal AI systems in production and research settings. Demonstrated record of delivering measurable results, 94% classification accuracy in real-time driver monitoring, 7 peer-reviewed publications, and 7 competitive awards. Deep expertise in computer vision, machine learning, signal processing, and XR-integrated AI pipelines. Proficient in Python, PyTorch, TensorFlow, and PySpark; hands-on with CNNs, LSTMs, Transformers, and LLM-powered agentic workflows. Adept at translating complex research into scalable, production-ready systems across healthcare, safety, and human factors domains. Incoming Postdoctoral Researcher at the University of Central Florida (UCF), Aug 2026.

TECHNICAL SKILLS

Languages	Python, R, SQL, MySQL, PostgreSQL
ML / DL	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, XGBoost, Gradient Boosting, CNN, LSTM, Transformer, ResCNN
Data Engineering	PySpark, Databricks, Pandas, NumPy, AWS (S3, SageMaker), Apache Spark
Computer Vision	OpenCV, YOLO Nano, MobileNet, ONNX, real-time inference pipelines
Signal Processing	EEG (Emotiv EPOC X), eye tracking, ECG/HR, feature extraction
XR / Hardware	HoloLens 2, Meta Quest Pro, Unity (OpenCV for Unity)
MLOps / Tools	Git, Linux, Docker (basic), LaTeX, MS Project, Excel (advanced dashboards)
AI / LLMs	LLM fine-tuning, Agentic AI, RAG pipelines, Prompt Engineering, NLP

PROFESSIONAL EXPERIENCE

Machine Learning & Computer Vision Researcher | *Mississippi State University — Dept. of Industrial & Systems Engineering*
Starkville, MS Aug 2022 – Aug 2026

- Built a real-time distracted driver detection system using PySpark and MobileNet on 2M+ frame video streams; achieved 94% classification accuracy — deployed as a proof-of-concept XR HMI via HoloLens 2.
- Designed multimodal cognitive workload classification pipelines fusing EEG (250 Hz, Emotiv EPOC X), eye-tracking, and heart rate data; engineered a 134-feature matrix and evaluated 9 classifiers — Gradient Boosting and ResCNN reached 94% accuracy (macro F1 = 0.93).
- Developed an AI-enabled immersive VR training system for geriatric nursing (NSF Grant #2222485); validated system usability and adoption via the Technology Acceptance Model (TAM) with gerontological nurses, yielding statistically significant improvements in training efficiency.
- Engineered a facial expression recognition pipeline on FER2013 (76% accuracy); integrated the model into a Unity-based XR interface using the ONNX runtime for on-device inference.
- Built a cost-analysis and decision-support Excel dashboard (DOE-sponsored) for industry partners John Deere and ZF Group to evaluate remanufacturing economics; streamlined data-driven planning workflows.
- Authored and co-authored 7 peer-reviewed journal articles and 4 conference papers; presented at INFORMS and IISE Annual; served as Session Chair for Image Processing & Computer Vision at IISE 2023.

EDUCATION

Ph.D., Industrial & Systems Engineering (Minor: Computer Science) Aug 2022 – Aug 2026
Mississippi State University, Starkville, MS
Dissertation: Integration of Artificial Intelligence and Extended Reality: Process, Implementation, Assessment, and Adoption

M.S., Industrial & Systems Engineering — Healthcare Systems Engineering

Sep 2017 – Jan 2019

*Tarbiat Modares University, Tehran, Iran***B.S., Industrial Engineering**

Sep 2012 – Jan 2016

*Shahid Bahonar University, Kerman, Iran***PEER-REVIEWED PUBLICATIONS**

Journal Articles

- Kashef, A., Wang, Y., & Ma, J. (Under Review). Real-Time Driver Distraction Monitoring via an AI-Powered Extended Reality: A Human-Machine Interface. *IEEE Transactions on Human-Machine Systems*.
- Kashef, A., Karwowski, W., & Ma, J. (Under Review). Multimodal Physiological Signal Fusion for Cognitive Workload Classification in Intelligent Immersive VR-Based Geriatric Nursing Training. *Computers in Human Behavior*.
- Kashef, A. et al. Developing a Novel AI-Enabled Extended Reality System for Real-Time Automatic Facial Expression Recognition and System Performance Evaluation. *Advanced Engineering Informatics*.
- Kashef, A. et al. Adoption Evaluation of an AI-Enabled Immersive Virtual Reality Tool in Gerontological Nursing Using an Extended Technology Acceptance Model. *International Journal of Industrial Ergonomics*.
- Kashef, A. et al. Treatment Outcome Classification of Pediatric Acute Lymphoblastic Leukemia Patients Using Machine Learning. *Informatics in Medicine Unlocked*.
- Kashef, A. et al. Prediction of Cranial Radiotherapy Treatment in Pediatric Acute Lymphoblastic Leukemia Using Machine Learning. *Asian Pacific Journal of Cancer Prevention*.
- Kashef, A. et al. Circular Economy Conceptualization Using Text Mining Analysis. *Sustainable Production and Consumption*.
- Kashef, A. et al. TRIZ-Based Design Improvement for Facilitating Transmission Control Unit Remanufacturing. *Journal of Mechanical Design*.

AWARDS & HONORS

- 1st Place Best Student Paper Award — Work Systems Division, IISE Annual Conference (May 2026)
- Best Publication Award — Dept. of ISE, Mississippi State University (May 2026)
- Outstanding Ph.D. Student Award — Dept. of ISE, Mississippi State University (May 2026)
- Paper of Distinction Award — IDETC/CIE Design for Manufacturing and Life Cycle Conference (Apr 2025)
- NSF IISE Student Travel Award (Jun 2025)
- Invention Contributor Recognition — University of Connecticut (Feb 2026)
- Bagley College of Engineering Graduate Assistantship Award (Jan 2024)

CERTIFICATIONS & TRAINING

- 5-Day AI Agents Intensive — Google
- Generative AI with Large Language Models — DeepLearning.AI & AWS
- Deep Learning Specialization — DeepLearning.AI
- Accelerated Computing with CUDA Python — NVIDIA
- Digital Signal Processing 1 — EPFL